

RANDY CHAN YI ZHONG

GenAI Solutions & Transformation Engineer

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EXECUTIVE SUMMARY

GenAI Solutions & Transformation Engineer with 7+ years turning business and workflow ambition into production AI systems across financial services and healthcare. Leads solutioning end to end (requirements workshops, solution architecture, and business-process / workflow automation), then delivers the engineering: multi-agent orchestration (LangGraph), enterprise RAG, model-serving optimization, and agentic governance. Cut a regulated credit-memo workflow from 6–24 hours of analyst time to 5–60 minutes on a system serving 10,000+ users at 100 percent uptime. ECBA-certified (IIBA) business-analyst who pairs deep technical depth (Python, AWS, Azure, Kubernetes) with the stakeholder-facing communication that gets systems trusted, adopted, and measurably faster, and drives team-wide AI-assisted engineering with Claude Code.

TRANSFORMATION HIGHLIGHTS

- **Cut credit-memo turnaround from 6–24 hours (human analysts) to 5–60 minutes** on a regulated, autonomous multi-agent workflow serving **10,000+ users** at **100 percent uptime** since launch.
- **Reduced inference latency by approximately 85 percent** (roughly 700–1,400 ms to 100–250 ms) via a concurrent orchestration layer for parallel tool-calling and asynchronous agentic flows.
- **Lifted operational productivity 50 percent** for a nation-wide healthcare system through Python-based backend automation (config-change tracking + validation).
- **Cut deployment time 40 percent** with Terraform + GitHub Actions CI/CD, and drove team-wide adoption of AI-assisted engineering (Claude Code) across development, design, and product workflows.

SKILLS

GenAI & LLMs: Multi-Agent Orchestration, LangGraph, LangChain, RAG Architectures, Model Fine-Tuning, LLM Evaluation (LangFuse, LLM-as-a-judge), Prompt Engineering, NLU, Intent Classification, Human-in-the-Loop / Agentic Governance, Responsible AI.

Solutions & Transformation: Solution Architecture, Requirements Analysis & Stakeholder Workshops, Business-Process / Workflow Automation, Process Optimization, BPMN, AI Adoption & Change Enablement, Stakeholder Management, Pre-sales / Solutioning.

Model Serving: OpenAI API, Anthropic API, HuggingFace, vLLM, PyTorch, Token Optimization, Latency & Throughput Tuning, Multi-provider Abstraction.

Backend: Python (FastAPI, Flask), Java (Spring Boot), NodeJS, REST API Design, OpenAPI / Swagger.

Frontend: React, VueJS, TypeScript, JavaScript (ES6+), HTML / CSS, Three.js (React-Three-Fiber), responsive SPA development.

Data & State: Vector DBs (Pinecone, Weaviate), MongoDB Atlas, PostgreSQL, Redis.

Cloud & Infra: AWS (ECR, ECS, S3), Azure (Container Apps, Service Bus, Blob Storage), Docker, Kubernetes, Terraform (IaC), CI/CD (GitHub Actions).

Ways of Working: AgentOps, MLOps, Agile / Scrum, Technical Documentation, Mentoring; Governance (PDPA, SOC 2, Responsible AI).

Certifications: Certified Associate AI Engineer (CAIE), AIP Singapore; Entry Certificate in Business Analysis (ECBA), IIBA.

EXPERIENCE

TrustPlus AI (Fintech)
Senior AI Software Engineer

Mar 2025 – Present

Project: Autonomous Credit-Risk & Multi-Agent Research Suite

- Cut credit-memo turnaround from 6–24 hours of analyst work to 5–60 minutes per memo by automating end-to-end credit-risk and brand-risk assessment for KYB, on a production multi-agent system (LangGraph, Python; 8 specialist research agents plus a supervisor) serving 10,000+ users at 100 percent uptime since launch, built in direct partnership with customer stakeholders.
- Reduced total inference latency by approximately 85 percent (roughly 700–1,400 ms to 100–250 ms) by designing a concurrent orchestration state layer for parallel tool-calling and asynchronous agentic flows, balancing model quality against throughput and token cost.
- Implemented agentic governance for responsible, compliant AI: Human-in-the-Loop review gates for high-risk credit decisions, configurable approval workflows, and an evaluator-optimizer loop (LLM-as-a-judge) that scores grounding against evidence and hard-fails contradicted claims.
- Established automated regression and evaluation pipelines (LLM-as-a-judge) validating tool-selection precision, factual faithfulness, and output drift across model versions; trained and evaluated NER and object-detection models to above 95 percent accuracy.
- Cut deployment time 40 percent by managing the full infrastructure lifecycle with Terraform (Azure Service Bus orchestrating 14 consumers, MongoDB Atlas, cloud-native containers) and GitHub Actions CI/CD; enforced SOC 2 and PDPA controls (PII anonymized before logging, encrypted at rest).
- Optimized RAG pipelines (semantic-aware chunking, reranking with reciprocal rank fusion, pre-caching) to minimize time-to-first-token for generated credit memos and financial-analysis reports across a multi-provider LLM abstraction (OpenAI, Anthropic, Google, Amazon Bedrock).
- Built an MCP server and an auto-generated Claude Code skill pack exposing the product's API (around 14 tool domains), and authored team-wide Anthropic Skills (code review, change planning, test review); the same tool layer is driven by Playwright in CI as an LLM-as-a-user end-to-end test on every deploy.

- Championed AI-assisted engineering across the team, driving adoption of Claude Code and establishing AI-augmented workflows that accelerated delivery while holding code-quality standards.

BGC Group

Nov 2024 – Mar 2025

Senior Software Engineer (AI Application Specialist), Contract

- Productionized RAG workflows for an enterprise Task Assistant in direct collaboration with business users; architected the full-stack application layer (FastAPI) for internal knowledge management.
- Engineered backend middleware orchestrating the application layer and AI model endpoints for high-concurrency, low-latency responses; served open-weights models with vLLM where predictable concurrency was cheaper per query than the API.
- Built stateful, multi-turn conversational sessions with Human-in-the-Loop escalation to human agents for complex intents; optimized retrieval precision with semantic-aware chunking and t-SNE embedding analysis.
- Integrated Weaviate (vector) and MongoDB (state) across heterogeneous stores; containerized CI/CD with Docker across development, staging, and production.

AI Singapore

Jan 2024 – Oct 2024

AI Engineer

- Architected an end-to-end production AI pipeline for a healthcare environment with clinical domain experts, integrating five specialized NER and event-prediction models with a focus on patient-data privacy; hyperparameter tuning yielded a 50 percent increase in model performance.
- Fine-tuned BioClinicalBERT for clinical NER (diseases, treatments, medications) served via FastAPI and orchestrated with a Kedro pipeline for reproducible training and eval; used the extracted entities as features for a downstream intervention-prediction classifier.
- Championed AI explainability with Explainable Boosting Machines (EBM) and interpretability frameworks so clinicians could read why each prediction was made, meeting medical transparency requirements.
- Prototyped a LangGraph-orchestrated multi-agent RAG research pilot for activity planning (FastAPI + Pinecone, deployed on AWS ECS/ECR); mentored 12 apprentices across two cohorts in NLP and LLM techniques.

NCS Pte. Ltd.

Oct 2022 – Dec 2023

Business Systems Analyst (Backend Automation & Integration)

- Improved operational productivity 50 percent for a nation-wide healthcare web application by engineering a Python automation suite to track and validate complex JSON configuration changes across distributed environments.
- Orchestrated end-to-end data-migration pipelines between SQL databases and virtualized environments, ensuring 99.9 percent data integrity for high-traffic production systems.
- Authored technical documentation and system / workflow diagrams (BPMN), communicating complex backend architecture to cross-functional teams and advising on PDPA regulatory compliance.

Bitopia Technology Pte. Ltd.

Jan 2022 – Oct 2022

Full-Stack Software Engineer

- Delivered scalable RESTful microservices for logistics and real-estate platforms, leading client-facing requirements workshops and pre-sales scoping to translate domain workflows into modular service architectures.
- Automated core logistics and real-estate workflows (digitizing manual dispatch, tracking, and listing/document handling) and integrated third-party services (such as mapping, carrier, and payment APIs) so operations ran end to end on one platform.
- Optimized PostgreSQL schemas and complex queries powering real-time logistics tracking, improving dashboard response times so clients could meet SLA requirements.

Napshot Pte. Ltd.

Jan 2019 – Dec 2021

Solutions & Software Engineer

- Led end-to-end solution design and delivery for SME clients across F&B, hospitality, and HR, serving as the primary technical point of contact for 10+ concurrent client engagements, running requirements workshops and translating business workflows into technical specifications.
- Digitized manual SME operations by replacing paper- and spreadsheet-based workflows with custom software across F&B, hospitality, and HR clients, and integrating disparate systems (point-of-sale, payments, third-party tools) into a single operational workflow per client.
- Built custom platforms (VueJS, ReactJS, Flask) and owned cloud infrastructure with 99.9 percent uptime; managed provisioning, monitoring, and incident response across multiple client deployments.

SELECTED PROJECTS

- **Leeseidon:** Private-first, end-to-end-encrypted investment-analytics SaaS (React, Cloudflare Workers/D1, Web Crypto AES-GCM + PBKDF2). Browser-side crypto with a ciphertext-only backend and a CLI runtime sharing the same compute as the web app; institutional analytics (TWR / MWR / Sharpe). Sole author across 185+ commits, with a two-secret model (account password vs. vault passphrase) and encrypted cross-device sync.
- **Jarvis:** Personal agentic “second brain” on Claude Code (Python, MCP, SQLite, on-device STT/TTS). Governance-first: human-approval execution gates, an immutable audit log, read-only finance by construction, and local voice; runs on a subscription model with no per-token API billing. Multi-surface (Telegram, terminal, on-device voice), integrating read-only IBKR/Longbridge brokers, Strava/Garmin health, and photo/voice calorie logging.

- **Kestrel MCP:** MCP-native policy-review tool for Singapore financial advisers (FastAPI, OAuth 2.1, Supabase). Compliance-by-design: the customer's own Claude reads documents and drafts advice while the backend stays a CRUD + validator + immutable-audit system of record, gated by mandatory human confirmation. PLG economics: near-zero inference cost since the customer brings their own Claude; per-field confidence with source citations.
- **Transaction Auto-Tagging Agent:** Month-end-close acceleration agent for finance teams (Python, React). Zero-silent-error by design: codes transactions against the tenant's chart of accounts, abstains when unsure, guards look-alike vendors, and improves via a per-tenant feedback loop with an append-only audit. Resolves unclear vendors via web-search-before-ask, with streaming, resumable runs and full event replay.
- **TWC:** Browser-only group-expense splitter with a multimodal chat assistant that reads receipts (React, multi-provider LLM + local Ollama). Agentic safety rails: a plan mode that strips mutating tools to resist prompt injection, Zod-validated tool calls, and bring-your-own-key spend caps. Fully client-side with a typed tool registry, slash commands, and streaming phase labels; no backend, no auth.
- **Verona:** Production wedding super-app PWA, live in production (Next.js, Supabase, Cloudflare R2/Workers). End-to-end ownership: server-enforced RSVP deadlines, gated CI/CD with contract-parity checks, and an OpenAPI parity contract that lets a bring-your-own-Claude agent drive the same surface as the UI, all on hard free-tier cost guardrails. Tri-lingual installable PWA (English, Chinese, Malay) with bulk CSV guest import, Web Push, and R2-backed media.

EDUCATION & CERTIFICATIONS

Singapore University of Social Sciences

B.Sc. in Information and Communication Technology (GPA 4.2/5) | Minor: Data Science

Certified Associate AI Engineer (CAIE), AI Professionals (AIP), Singapore · Entry Certificate in Business Analysis (ECBA), IIBA.